

PROJECT REPORT OF CO-OP Project at Industry (Module-2)
ON
Mood Detection and Environment Analyzation
submitted in partial fulfilment of the requirements for the award of degree of
BACHELOR OF ENGINEERING
In
COMPUTER SCIENCE AND ENGINEERING

Submitted by:
Rupali Wadhwa
2210990748

Supervised By:
Dr. Gurpreet Singh
Associate Professor
Chitkara University



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY
CHITKARA UNIVERSITY, PUNJAB, INDIA

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **Mood Detection and Environment Analyzation**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Gurpreet Singh**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Date: 17/05/2026

Rupali Wadhwa

(University Roll No.-2210990748)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

Dr. Gurpreet Singh

Associate Professor

Department of Computer Science and Engineering

Chitkara University Institute of Engineering and Technology,

Chitkara University, Punjab, India

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We are grateful for his help which played a very foremost part in the project and for providing us with all the indispensable information for developing this ML application. I am profoundly grateful to each individual and organization mentioned above, as well as to all those who, in various capacities, have contributed to the realization of this project. Your support, guidance, and encouragement have been invaluable, and I am truly honored to have had the opportunity to work with such remarkable individuals.

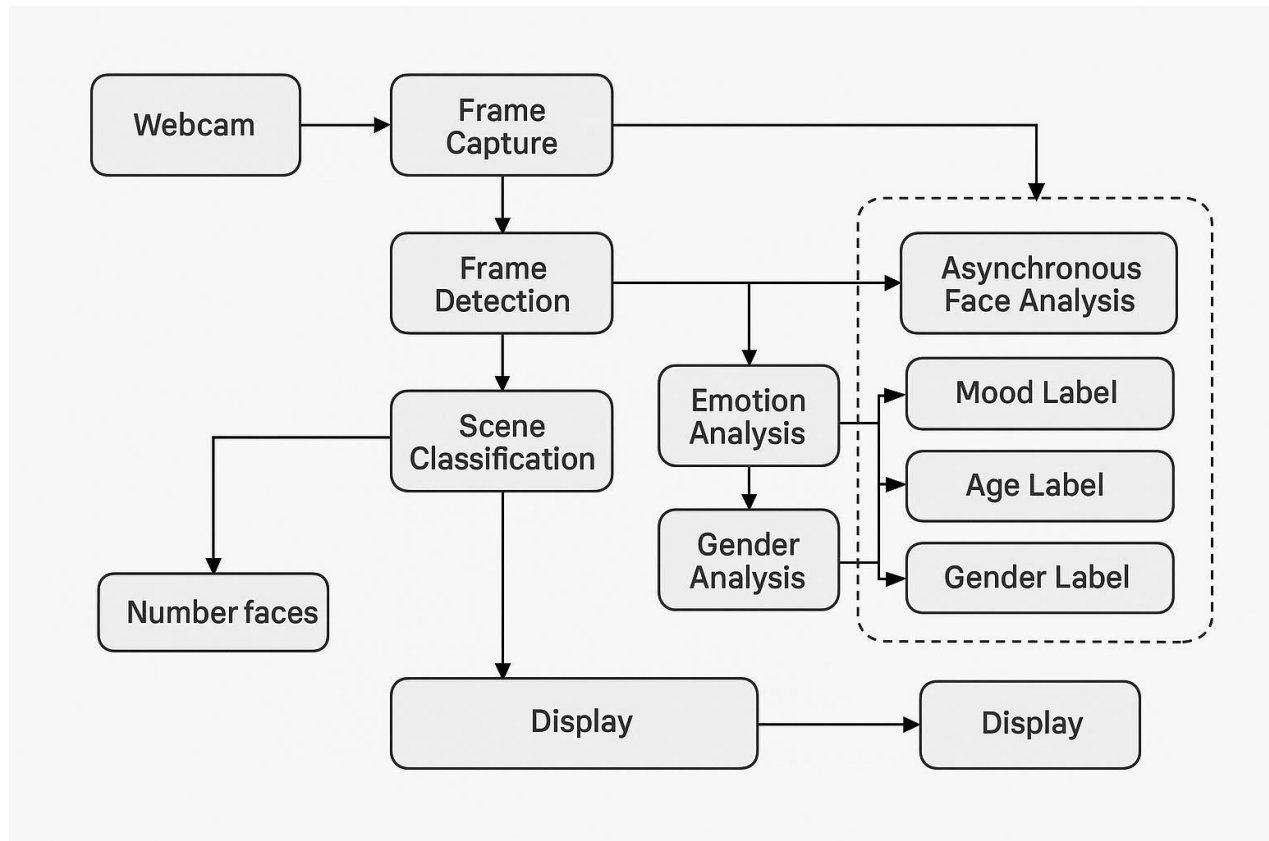
Rupali Wadhwa
(2210990748)

LIST OF FIGURES AND TABLES

1. Comparative Study

Feature	Existing Systems	Proposed System
Data Requirement	Requires training datasets	No dataset required
Emotion Model Training	Required before deployment	Not required
Context Awareness	Mostly absent	Includes environment recognition
Deployment Type	Often offline or semi-live	Fully real-time
Multi-Person Detection	Limited	Supported
Scalability	Moderate	High – modular design
Processing Method	Single-threaded or static	Multithreaded, dynamic inference
Visualization	Minimal	Overlays + dynamic emotion graphs

2. System Flow Diagram



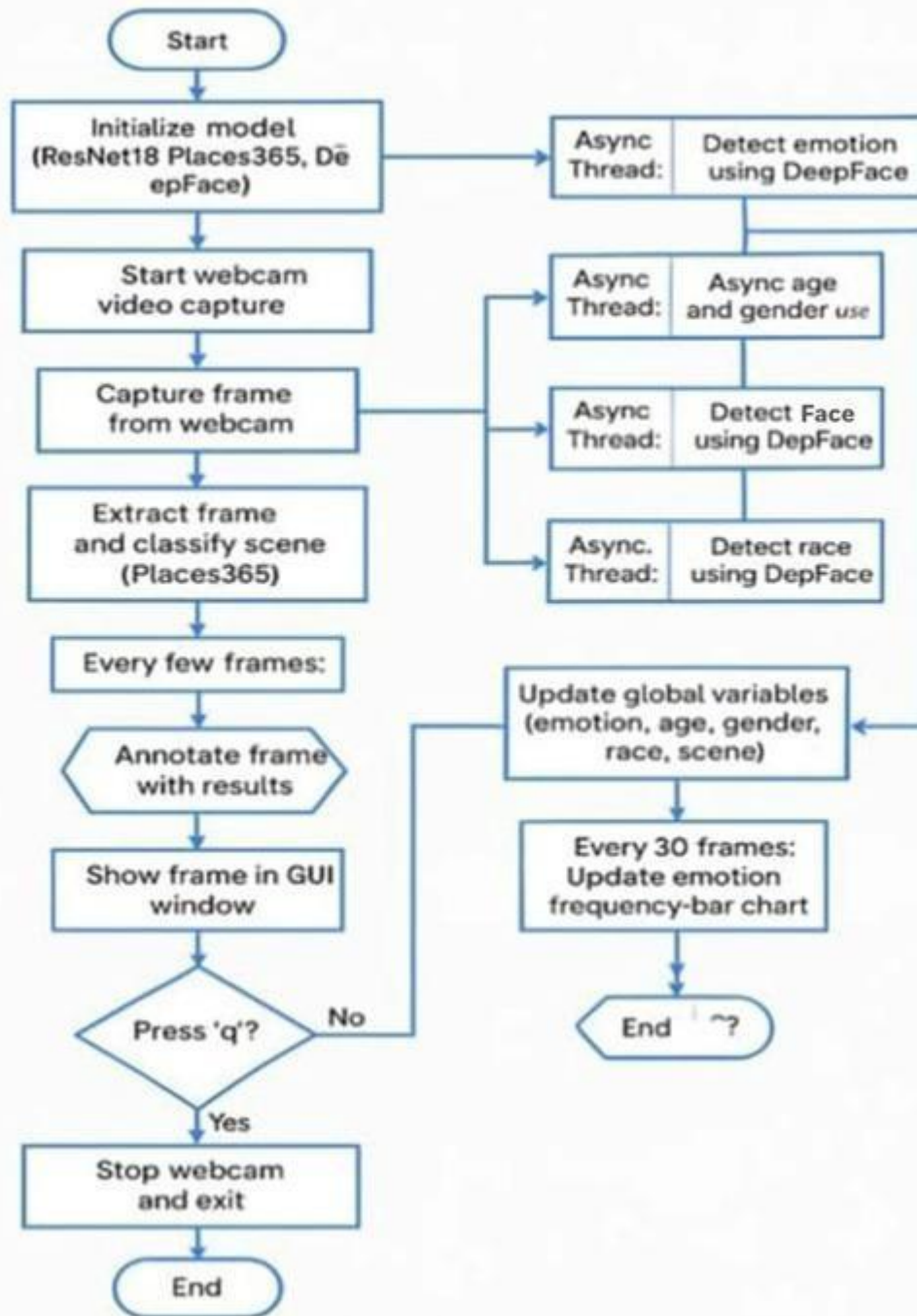
3. System Modules

Module Name	Description
Live Video Capture Module	Captures real-time frames from the webcam using OpenCV.
Preprocessing Module	Converts input into optimal format for model inference (RGB conversion, optional resizing).
Mood Detection Module	Uses pretrained DeepFace inference model to classify emotional state from each frame.
Environment Recognition Module	Utilizes pretrained Places365 to determine the environment classification.
Aggregation & Analytics Module	Tracks emotion frequency, number of individuals detected, and time-based patterns.
Visualization Module	Displays live detection results, overlays, person count, and emotional graph.
Control/Exit Module	Allows the user to terminate the application safely (e.g., using key input).

4. System Architecture Design

A Hybrid Deep Learning Framework for Simultaneous Mood and Environment Detection

A Hybrid Deep Learning Framework for Simultaneous Mod and Enon/x



ABSTRACT

Human behavior analysis is becoming increasingly essential across domains such as security, healthcare, retail, and personalized AI-driven systems. This project presents a real-time multimodal facial and environmental analysis system capable of detecting emotion, age group, gender, race, number of faces, and surrounding scene context using a live webcam feed. The system integrates computer vision, deep learning, and pre trained models to achieve efficient and accurate predictions.

Using the DeepFace framework, the system extracts and interprets facial attributes, while OpenCV is used for face detection, frame capturing, and visual annotation. In parallel, a pre-trained Places365 ResNet18 model is employed to analyze the environment and classify the surrounding scene. The results are processed asynchronously using multithreading to ensure smooth performance without compromising detection accuracy.

The system provides a combined and interactive output overlaying predictions onto a live video stream, enabling real-time monitoring and analytics. This approach demonstrates how AI-powered behavioral analysis can be utilized in real-world applications including smart surveillance, adaptive user interfaces, psychological studies, audience analysis, and human–computer interaction.

The project successfully proves that integrating multiple AI frameworks can enhance contextual awareness and provide meaningful insights into both individuals and their environments in real time.

Key Features:

1. Real-Time Mood Recognition

The system continuously analyzes live video input to detect and classify emotional states such as happiness, sadness, anger, fear, surprise, and neutrality, providing instant feedback with high accuracy.

2. Context-Aware Environment Classification

A pre-trained ResNet18 Places365 model is used to identify the surrounding environment (e.g., office, cafeteria, corridor, classroom), allowing emotional patterns to be interpreted with contextual intelligence.

3. Multi-Person Detection and Mood Aggregation

The model is capable of recognizing the presence of multiple individuals in a frame and can aggregate emotional responses, enabling applications in group environments such as classrooms, meetings, or public spaces.

4. High-Performance Asynchronous Processing

The system uses a multithreaded architecture to run mood classification, scene understanding, object analysis, and live display simultaneously without interrupting the video stream or reducing frame rate.

5. Real-Time Visual Output Overlay

Detected results such as mood labels, scene descriptions, detected person count, and timestamps are displayed directly on the live video feed, ensuring intuitive and instant analysis.

6. Live Emotion Trend Visualization

A dynamic graph updates periodically to represent the frequency of detected emotions over time, enabling behavioral pattern tracking suitable for research, mental wellness monitoring, and human-computer interaction studies.

7. Timestamped Behavioral Logging

The system embeds date and time information into the stream and can be extended to store time-based emotional patterns for longitudinal behavioral observations or analytic reports.

8. Scalable and Modular Design

The architecture is designed to support future extensions, such as integrating sentiment-based voice analysis, database logging, or advanced behavioral analytics, without modifying the core structure.

9. User-Friendly Interface

The system operates through a simple and interactive interface where results are displayed clearly without requiring technical knowledge, making it accessible for academic, professional, and real-world deployment.

1. INTRODUCTION

Human–computer interaction is rapidly evolving through systems that can interpret human behavior and respond intelligently. Mood detection using artificial intelligence has become an important research direction due to its applications in mental health analysis, adaptive learning, marketing analytics, security surveillance, and personalized computing. Traditionally, computers respond only to explicit commands, but recent advancements in deep learning and behavioral computing enable systems to understand non-verbal cues such as emotional state, behavior trends, and contextual surroundings.

In parallel, environmental understanding plays a key role in interpreting human responses more accurately. Emotions often vary with surroundings—for example, individuals may exhibit calmness in a library, alertness in a classroom, or excitement in recreational spaces. Studies in affective computing also emphasize that combining emotional signals with environmental context significantly improves decision-making accuracy.

This project integrates real-time mood detection with scene/environment recognition, creating a combined intelligent system capable of identifying emotional states in context. The system uses live video input and pre-trained deep learning models to analyze mood patterns, detect multiple persons simultaneously, classify surroundings, and visualize emotional frequency over time. The outcome demonstrates the potential of context-aware emotional AI for real-world applications.

1.1. Problem Outline

Although several systems exist for emotion detection, most solutions focus only on identifying emotional states without considering the surrounding conditions in which the emotion occurs.

This creates challenges such as:

- Lack of contextual understanding when analyzing emotions.
- Difficulty interpreting emotional changes influenced by environment or group dynamics.
- Limited scalability for real-time or multi-person environments.
- Absence of visualized tracking for emotion trends and behavioral patterns.

Existing research papers highlight that mood recognition systems lose accuracy when they ignore external factors such as lighting, background, noise, or location context. Therefore, there is a need to develop an integrated system that detects emotional states along with environmental cues to achieve meaningful human behavior interpretation.

1.2.Problem Objective

The primary objective of this project is to design and develop a real-time intelligent system that:

- Detects and classifies emotional states from live video streams using deep learning.
- Identifies and analyzes the surroundings to provide environmental context.
- Supports multi-person emotional recognition and visualization simultaneously.
- Displays emotional states, scene information, timestamps, and person count in real time.
- Generates emotion frequency graphs dynamically to track behavioral trends.
- Demonstrates scalable architecture suitable for research, behavioral analytics, and real-world deployment.

1.3.Limitations

Although the system performs efficiently, there are some limitations:

- Accuracy may be affected by low lighting, extreme camera angles, or motion blur.
- The system depends on pre-trained models, which may not represent cultural or demographic diversity perfectly.
- Emotional recognition is limited to predefined classes and may not detect mixed or subtle mood states.
- The system currently does not include long-term storage or cloud-based analytics, although the architecture allows future integration.
- Scene recognition may sometimes misclassify visually similar environments.

1.4.Summary

This project presents a real-time mood detection and environment recognition system that bridges the gap between emotional AI and contextual awareness. By integrating multiple deep learning models, asynchronous processing, and dynamic visualization, the system provides meaningful insights that can support mental health monitoring, workplace analytics, intelligent learning platforms, and adaptive user interfaces. Though certain limitations exist, this work establishes a strong foundation for future development in **affective computing, behavioral analytics, and intelligent adaptive environments.**

2. METHODOLOGY

The system follows a structured step-by-step pipeline:

1. Capturing live video stream using the webcam.
2. Processing frames for emotion inference using pretrained models.
3. Running environment analysis at intervals for context recognition.
4. Detecting and counting individuals present in the frame.
5. Annotating the results directly onto the live feed.
6. Updating the real-time emotion distribution graph.
7. Displaying combined output through an interactive window.

3. TOOLS AND TECHNOLOGIES

In this project, various modern tools and technologies have been used to develop a real-time mood detection and environment analysis system. The focus was on using reliable frameworks and pre-trained models to achieve accurate results while maintaining smooth and efficient performance.

Key tools and technologies used are:

- **Python** – Used as the main programming language for developing the entire system.
- **OpenCV** – Helps in capturing live video, detecting faces, and displaying results with proper visual output.
- **DeepFace** – Used for detecting facial attributes such as emotion, age, gender, and race.
- **ResNet18 (Places365 Model)** – A pre-trained model used to recognize and classify the surrounding environment.
- **MTCNN** – Used for accurate face detection and identifying facial landmarks.
- **Deep Learning & CNNs** – Core techniques used for processing and analyzing image data.
- **Computer Vision** – Enables the system to understand and interpret visual input from the webcam.
- **Multithreading** – Ensures smooth execution by running multiple tasks simultaneously without slowing down the system.

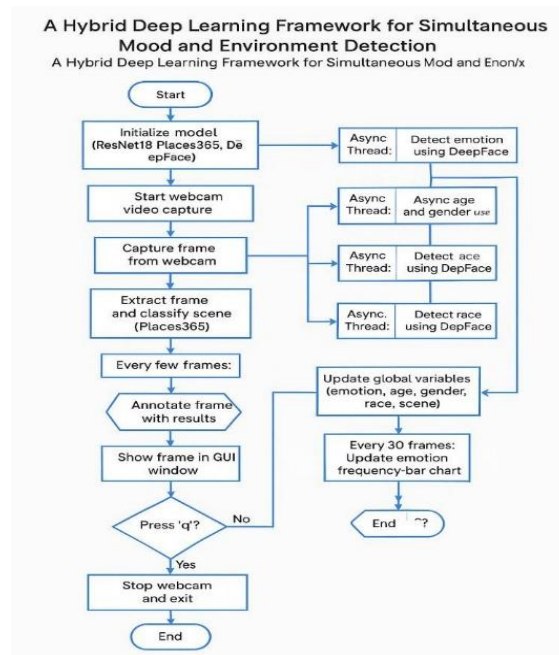
4. IMPLEMENTATION

The implementation of this system is based on a real-time workflow where the webcam is used to continuously capture video frames. Each frame is processed using OpenCV, and facial features are analyzed using the DeepFace model to detect emotions, age, and gender. At the same time, the Places365 model is used at intervals to identify the surrounding environment. The system is designed to handle multiple faces in a single frame and display the detected results such as mood, environment, and person count directly on the screen. To ensure smooth performance, multithreading is used so that all tasks run simultaneously without any delay. Overall, the system provides accurate and real-time output through an interactive interface.

4.1. Implementation Workflow

The implementation consists of the following steps:

1. Initialize the webcam.
2. Capture and read frames from the camera in a continuous loop.
3. Process frames to ensure they are compatible with the required model format.
4. Run mood detection using the DeepFace inference model.
5. Run environment analysis using the Places365 model at scheduled intervals.
6. Detect multiple individuals and apply aggregation logic.
7. Render output labels, bounding overlays, and graphs.
8. Continue processing until the user terminates the application.



5. RESULTS

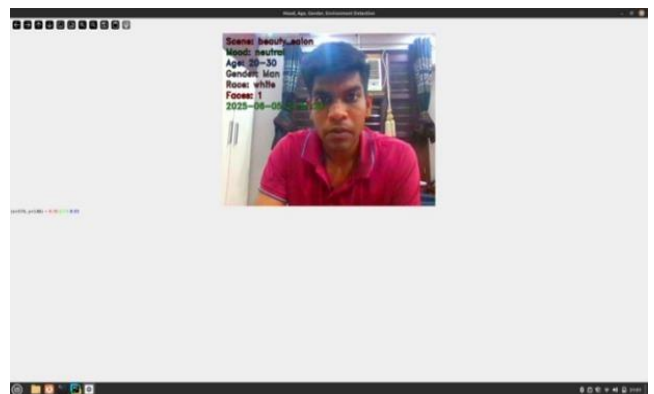
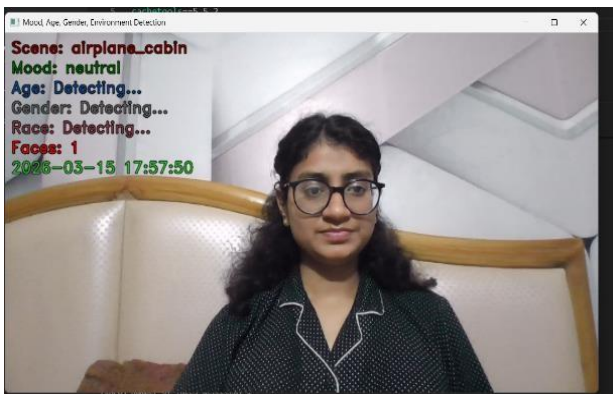
5.1.Introduction

This chapter presents the outcomes of the implemented real-time mood detection and environment recognition system. The results are evaluated based on system performance, detection accuracy, responsiveness, and user interaction experience. The discussion further explains how the system meets the project objectives and how its behavior compares with expectations and existing approaches.

5.2.Output Sample

During execution, the system successfully performed the following tasks in real time:

- Detection of mood categories such as happy, neutral, sad, angry, surprised, disgusted, and fearful.
- Identification of the environment label such as office, classroom, corridor, cafeteria, or outdoor space using the pretrained Places365 model.
- Display of real-time overlays including:
 - o Detected emotion label(s)
 - o Environment type
 - o Timestamp
 - o Number of individuals detected
 - o Live emotion frequency updates



A graphical representation of emotional trends was displayed alongside the live video feed to provide a visual summary of emotional patterns over time

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The development of the **Real-Time Mood Detection and Environment Analysis System** demonstrates how artificial intelligence can be integrated into live systems to interpret human emotional response with contextual awareness. Unlike traditional approaches that rely on curated datasets and offline processing, this system utilizes pretrained models to analyze emotions and environment using real-time webcam input. This eliminates dataset dependency and makes the solution accessible, scalable, and deployable in everyday settings.

The system successfully performs real-time emotion identification, multi-person awareness, and scene classification while providing visual and analytical representations through live output overlays and trend graphs. The integration of environment recognition enhances emotional interpretation, creating a more meaningful behavioral analysis framework.

Overall, the system achieved its intended goals of providing an intelligent, responsive, and context driven mood recognition platform suitable for applications in education, behavioral analytics, security, adaptive user interfaces, and mental health support.

6.2 Future Scope

Although the current system demonstrates effective performance, there are several areas where enhancements can be made to strengthen functionality and broaden applicability. Potential future improvements include:

- **Advanced Emotion Spectrum Recognition:** Expanding emotion detection beyond basic categories to include subtle emotions such as confusion, boredom, confidence, and mixed emotional states.
- **Voice-Based Sentiment Integration:** Combining audio analysis with visual mood detection for improved accuracy in real-time emotional inference.

- **Cloud-Based Processing and Storage:** Adding remote logging and cloud dashboard features for long-term emotional tracking, reporting, and professional use cases.
- **Mobile and IoT Deployment:** Creating lightweight models suitable for smartphones, smart mirrors, smart classrooms, and embedded systems.
- **Facial Landmark and Body Language Interpretation:** Including gesture recognition, engagement monitoring, and behavioral cues such as posture and gaze tracking.
- **AI-Based Learning Enhancements:** Allowing model refinement using user-specific feedback, enabling personalization over time.
- **Cross-Cultural Model Adaptation:** Improving recognition accuracy across diverse demographic, cultural, and environmental variations. 12 With these enhancements, the system can evolve into a robust emotional intelligence platform for commercial, academic, and healthcare environments.

6.3 Summary

This chapter summarized the outcomes, strengths, and technical contributions of the proposed system. The project successfully bridges the gap between traditional facial emotion recognition and context aware behavioral AI. The identified future improvements offer a strong roadmap for expanding this work into a versatile, intelligent monitoring solution.

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APPENDICES

Appendix A – Tools and Technologies Used

- Python
- OpenCV
- DeepFace
- ResNet18 (Places365)
- PyTorch
- Matplotlib
- Multithreading

Appendix B – Models and Framework Details

- DeepFace for emotion, age, gender, and race detection
- ResNet18 for scene classification
- OpenCV for webcam processing and visualization
- PyTorch for deep learning model execution

Appendix C – Workflow Summary

- Webcam input collection
- Face detection
- Emotion analysis
- Scene recognition
- Real-time output display
- Graphical emotion visualization

Appendix D – System Requirements

- Python environment
- Webcam support
- Minimum RAM and processor requirements
- Required libraries and dependencies

Appendix E – Additional Project Information

- Real-time processing approach
- Multithreading implementation
- Future enhancement possibilities
- Practical applications of the system